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# Column transformer

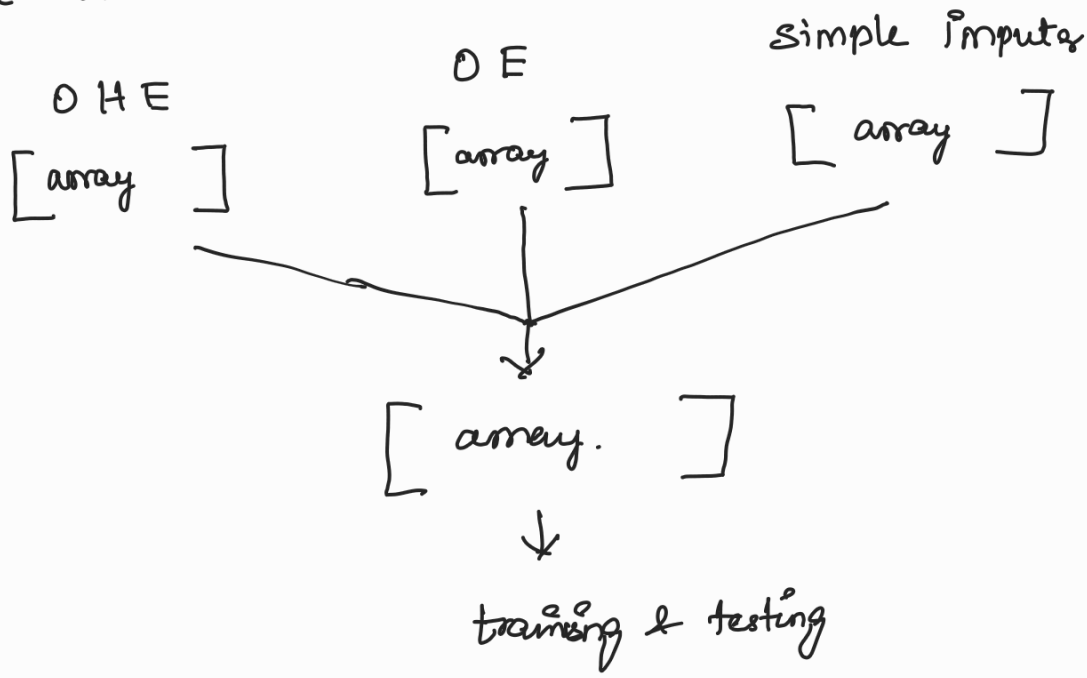
While handling the categorical data, we saw different types like Nominal and ordinal categorical data. For Nominal categorical data we used One hot encoder method. And for ordinal categorical data we used ordinal encoder.

For a missing value in a dataset is resolved using a method called simple imputer. which replaces the space with mean value.

## Example :

| Student - ID | Age | Gender | Grade | Result |
|---|---|---|---|---|
| 0 | 18 | Male | HS | Good |
| 1 | 20 | Female | UG | Good |
| 2 | 22 | Male | PG | Poor |
| 3 | 21 | Male | PG | Average |
| 4 | 19 | Female | HS | Poor |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| | ⇓ | ⇓ | ⇓ | ⇓ |
| | found a Missing Values | Nominal category | Ordinal Category | ordinal category |
| | ↓ | ↓ | ↓ | ↓ |
| | Simple imputer | OHE | OE | OE |

In general case, we apply OHE, OE, simple imputer seperately and then at end we combine all to load inside train and test set.




This is a big + time consuming Process, to overcome this, we use column transformer method. Here we do all operation in one single step to get the result.

Code:

```
from sklearn.compose import ColumnTransformer

transformer = columnTransformer(transformer =
    [ ('t1', simpleImputer(), ['Age']),
      ('t2', OrdinalEncoder(categories = [['HS','UG','PG']],
                                          ['Grade'])
      ('t3', OrdinalEncoder(Categories = [['Poor','Average','Good']],
                                          ['Result'])
      ('t4', OneHotEncoder(sparse = False, drop='first'),
                                          ['gender']]],
      remainder='Pass through')
```